FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2024

TEST DATE: 17-12-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A hypothetical radioactive substance decays according to $N = N_0 - kt^2$ where N is instantaneous number of active nuclei and N_0 is initial number of active nuclei and k is a positive constant of appropriate dimensions. Select correct options. $\left[t \le \sqrt{\frac{N_0}{2}}\right]$

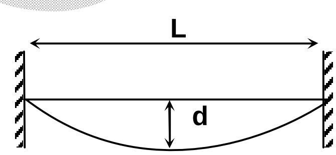
(A) Rate of decay when number of active nuclei becomes $\frac{N_0}{2}$ is $\sqrt{kN_0}$

(B) Rate of decay when number of active nuclei becomes $\frac{N_0}{2}$ is $\sqrt{2kN_0}$

(C) Rate of decay when number of active nuclei becomes $\frac{N_0}{2}$ is $\sqrt{\frac{kN_0}{2}}$

(D) Rate of decay when number of active nuclei becomes $\frac{N_0}{2}$ is $\sqrt{3kN_0}$

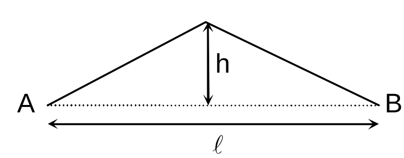
2. A wire having mass per unit length μ and length L is fixed between two fixed vertical walls at a separation L. Due to its own weight the wire sags. The sag in the middle is d(<<L). Assume that tension is practically constant along the wire, owing its small mass. Calculate the speed of the transverse wave on the wire.



(A) $L\sqrt{\frac{g}{4d}}$ (B) $L\sqrt{\frac{g}{8d}}$

(C) $L\sqrt{\frac{g}{2d}}$ (D) $L\sqrt{\frac{g}{16d}}$

3. Consider an elastic string stretched between two fixed ends. The string is plucked in the middle and is held in a triangular form with a maximum height h << ℓ at its middle point. At time t = 0, the plucked string is released from rest. All effects due to gravity may be neglected. If linear mass density of the string is μ and the speed of propagation for transverse waves in the string is 'c', find the total mechanical energy of vibrating string. (Assume that tension in the string does not change during oscillation)



(A) $\frac{\mu c^2 h^2}{\ell}$ (B) $\frac{\mu c^2 h^2}{2\ell}$

(C) $\frac{2\mu c^2 h^2}{\ell}$ (D) $\frac{\mu c^2 h^2}{4\ell}$

4. A small ball of density $4\rho_0$ is left from the surface of liquid. The density of liquid varies as $\rho = \rho_0(1+ay)$, where y is depth of liquid from the surface and ρ_0 & a are constants. Find the time period of oscillation of ball from its equilibrium position. $\left[a = 1.6\ m^{-1}, g = 10\ ms^{-2}\right]$. Neglect viscosity.
(A) π sec
(B) $\pi/2$ sec
(C) 2π sec
(D) $\pi/3$ sec

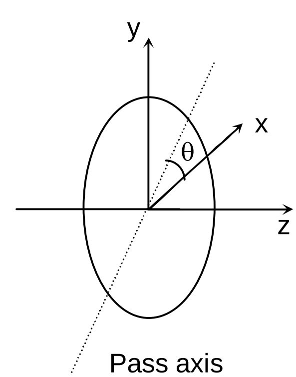
5. A standing wave given by the equation $y = 2A\sin\left(\frac{\pi x}{L}\right)\sin\omega t$ is formed between the points x = –L and x = L. What is the minimum separation between two particles whose maximum velocity is half the maximum velocity of the antinodes?
(A) $\frac{L}{12}$
(B) $\frac{L}{6}$
(C) $\frac{L}{3}$
(D) $\frac{2L}{3}$

6. An object is placed at a distance of 20 cm from an equi-concave lens of focal length 30 cm whose one side is silvered. If a virtual image is formed at $\frac{20}{3}$ cm from the lens then what is the radius of curvature of the curved surface?
(A) 30 cm
(B) 40 cm
(C) 60 cm
(D) 15 cm

7. A parallel beam of light of intensity I is incident on a perfectly absorbing disc of radius a. The beam is directed along the axis of disc. What is the total radiation force experienced by the disc.
(A) $\frac{2\pi a^2 I}{C}$
(B) $\frac{3\pi a^2 I}{C}$
(C) $\frac{\pi a^2 I}{C}$
(D) $\frac{\pi a^2 I}{2C}$

8. In a compound microscope, the focal length of objective lens is 1.2 cm and focal length of eye piece is 3.0 cm. When object is kept at 1.25 cm in front of objective, final image is formed at infinity. Magnifying power of the compound microscope should be:
(A) 200
(B) 100
(C) 400
(D) 150

9. A plane polarized light is incident on a polariser with its pass axis making angle θ with x-axis, as shown in the figure. At four different values of θ, θ = 9°, 39°, 189° and 219°, the observed intensities are same. What is the angle between the direction of plane of vibration of incident light and x-axis
(A) 204°
(B) 45°
(C) 98°
(D) 128°



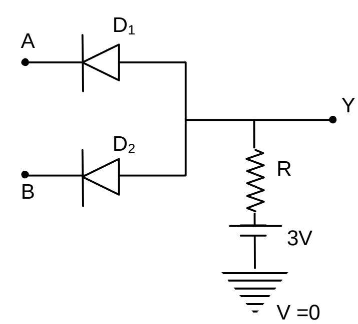
10. The pitch and the number of divisions, on the circular scale for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 division below the mean line. The readings of the main scale and the circular scale, for a thin sheet, are 4.5 mm and 48 respectively, the thickness of the sheet is:
(A) 4.755 mm (B) 4.950 mm
(C) 4.725 mm (D) 4.740 mm

11. Calculate of dimension of $\left(\frac{B^2}{\mu_0}\right)$. (where μ_0 : permeability of free space and B : magnetic field)
(A) $\left[ML^2T^{-2}\right]$ (B) $\left[MLT^{-2}\right]$
(C) $\left[ML^{-1}T^{-2}\right]$ (D) $\left[ML^2T^{-2}A^{-1}\right]$

12. A body of mass m is projected with velocity nv_e in vertically upward direction from the surface of the earth into space. It is given that v_e is escape velocity and n<1. If air resistance is considered to the negligible, then the maximum height from the centre of earth, to which the body can go, will be (R : radius of earth)
(A) $\frac{R}{1+n^2}$ (B) $\frac{R}{1-n^2}$
(C) $\frac{R}{1-n}$ (D) $\frac{n^2R}{1+n^2}$

13. The mass of an oxygen molecule is about 16 times that of a hydrogen molecule. At room temperature the rms speed of oxygen molecules is v. The rms speed of the hydrogen molecule at the same temperature will be :
(A) v/16 (B) v/4
(C) 4v (D) 16v

14. In the circuit, the logical value of A = 1 or B = 1 when potential at A or B is 3V and the logical value of A = 0 or B = 0 when potential at A or B is 0 V.



The truth table of the given circuit will be:

(A)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

(B)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

(C)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

(D)

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

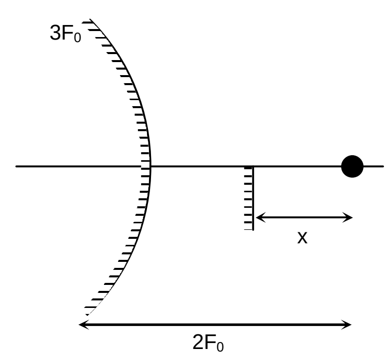
15. Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits in two fragments 'B' and 'C' of mass numbers 104 and 116. The binding energy of nucleons in 'B' and 'C' is 6.4 MeV per nucleon. The energy Q released per fission will be :
(A) 0.8 MeV
(B) 275 MeV
(C) 220 MeV
(D) 176 MeV

16. A plane electromagnetic wave travels in a medium of relative permeability 1.6 and relative permittivity 6.44. If magnitude of magnetic intensity is 4.5×10^{-2} Am^{-1} at a point, what will be the approximate magnitude of electric field intensity at that point? (Given: permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}, speed of light in vacuum $c = 3 \times 10^8$ m/s)
(A) 16.9 V/m
(B) 2.25×10^{-2} V/m
(C) 8.48 V/m
(D) 4.2×10^6 V/m

17. The relation between root mean square speed (V_{rms}) and most probable speed (v_P) for the molar mass M of oxygen gas molecule at the temperature of 300 K will be
(A) $v_{rms} = \sqrt{\frac{2}{3}}v_P$
(B) $v_{rms} = \sqrt{\frac{3}{2}}v_P$
(C) $v_{rms} = v_P$
(D) (A) $v_{rms} = \sqrt{\frac{1}{3}}v_P$

18. Two coherent sources of light interfere. The intensity ratio of two sources is 1 : 9 . For this interference pattern find the value of $\frac{I_{max} + I_{min}}{I_{max} - I_{min}}$
(A) 4/5
(B) 5/4
(C) 5/3
(D) 3/5

19. From a convex mirror of focal length $3F_0$ an object is placed at a distance of $2F_0$ as shown in the figure. Where should we place a plane mirror in between object and convex mirror so that image formed by both the mirrors is formed at the same location. Find the distance of plane mirror from object.



(A) $\frac{8F_0}{5}$
(B) $\frac{4F_0}{5}$
(C) $\frac{5F_0}{4}$
(D) $\frac{5F_0}{8}$

20. Two identical photocathodes receive light of frequencies f_1 and f_2. If the maximum velocities of the photoelectrons (of mass m) coming out are respectively v_1 and v_2, then :

(A) $v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

(B) $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2)\right]^{1/2}$

(C) $v_1^2 + v_2^2 = \frac{2h}{m}(f_1 + f_2)$

(D) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2)\right]^{1/2}$

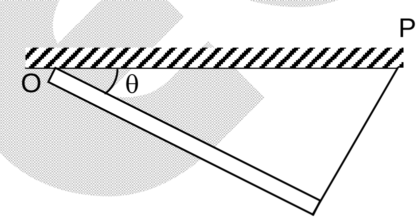
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

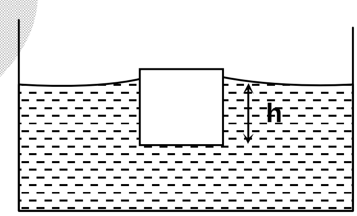
21. A radioactive sample decays through two different decay processes α and β. Half life time for α decay is 3 hr. and half life time for β decay is 6 hr. The ratio of number of initial radioactive nuclei to the number of radioactive nuclei present after 6 hr is....

22. A uniform rod of mass m and length L is hinged at one of its end with the ceiling and other end of the rod is attached with a thread which is attached with horizontal ceiling at point P. If one end of the rod is slightly displaced horizontally and perpendicular to the rod and released. The time period of small oscillation is $2\pi\sqrt{\frac{2\ell \sin\theta}{xg}}$. Find x.

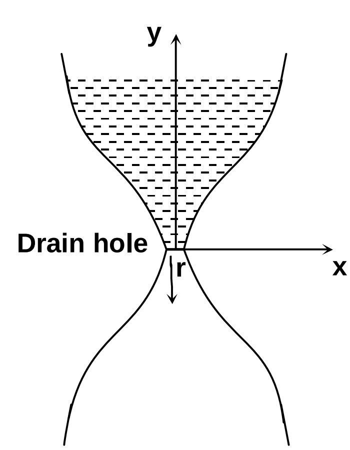


23. A cube of side a and mass m just floats on the surface of water as shown. The surface tension and density of water are S and ρ_w respectively. If angle of contact between cube and water surface is zero. If h is the distance between lower face of cube and surface of the water. Find $\left(\frac{h}{2}\right)$(in meter) between the lower face of cube and surface of the water.

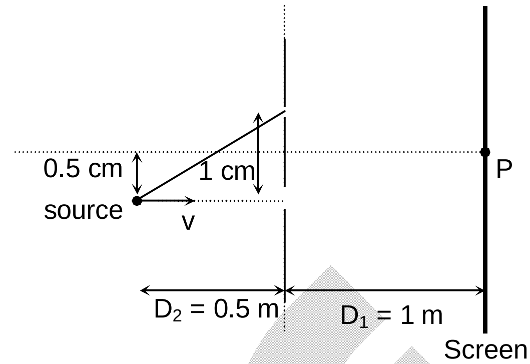
(Take $m = 1\,kg, g = 10\ m/s^2, Sa = \frac{10}{4}$ SI unit and $\rho_w a^2 g = 10$ SI unit)



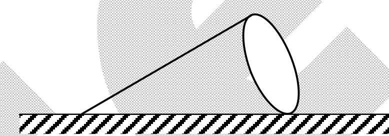
24. The shape of an ancient water clock jar is such that the water level descends at a constant rate at all times. The water level falls by λm/s and the shape of the jar is given by $y \alpha x^n$. Find the value of n if the radius r of the drain hole can be assumed to be very small.



25. In a typical Young's double slit experiment a point source of monochromatic light is kept as shown in the figure. If the source is given an instantaneous velocity v = 1 mm/s towards the screen, then the instantaneous velocity of central maxima is given as $\alpha \times 10^{-3}$ cm/s upward in scientific notation. Find the value of α.



26. A completely fluid filled hollow closed cone of height H and radius R is lying on the horizontal ground with the slant surface horizontal and taking the minimum pressure of the fluid to be zero. The net force on the flat surface of the cone by the fluid is $(10^4\sqrt{x})$ N. (Take the density of the fluid to be 1000 kg/m^3 and g = 10 m/sec^2, R = 1 m, H = 1 m, $\pi = \sqrt{10}$). Find x.



27. The potential energy of a particle of mass m is given by $V(x) = \begin{cases} E_0 & 0 \le x \le 1 \\ 0 & x > 1 \end{cases}$ λ_1 and λ_2 are the de-Broglie wavelengths of the particle, when $0 \le x \le 1$ and $x > 1$ respectively. If the total energy of particle is $2E_0$, find $\left(\frac{\lambda_1}{\lambda_2}\right)^2$.

28. A hydrogen like atom of atomic number z is in excited state of quantum number 2n. It can emit a maximum energy photon of 204 eV. If it makes a transition to the quantum state n, a photon of energy 40.8 eV is emitted. Calculate the atomic number z. Ground state energy of hydrogen atom is –13.6 eV.

29. When a certain metallic surface is illuminated with mono chromatic light of wavelength λ, the stopping potential for photoelectric current is $3V_0$. When the same surface is illuminated with light of wavelength 2λ the stopping potential is V_0. The threshold wavelength for this surface for photoelectric effect is $X\lambda$. Find X.

30. One mole of a mono-atomic gas is heated in such a way that its molar specific heat in the process is 2R. During the heating, the volume of the gas is doubled. T_i and T_f are the absolute temperatures of the gas at initial and final states respectively. Find the ratio of temperatures $\frac{T_f}{T_i}$.

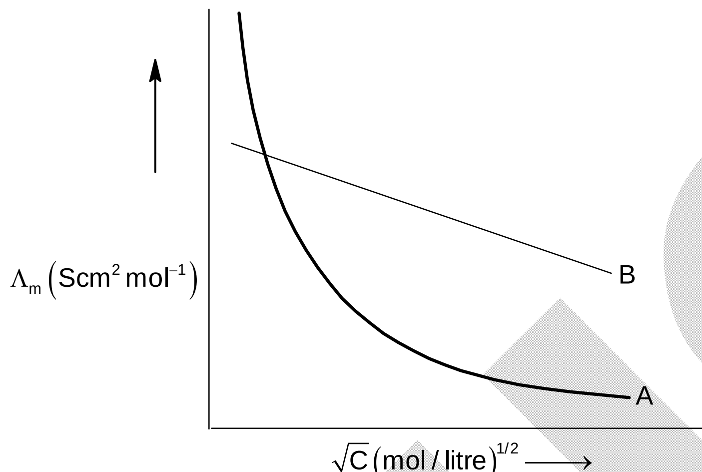
Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Following figure shows dependence of molar conductance of two electrolytes on concentration. Λ_m^o is the limiting molar conductivity.



Which one of the following statements is **incorrect** regarding above graph?
(A) A represents the molar conductance's variation with concentration for weak electrolyte like acetic acid.
(B) B represents the molar conductance's variation with concentration for strong electrolyte like potassium chloride.
(C) For weak electrolyte Λ_m^o can not be obtained by extrapolation of graph.
(D) Λ_m^o for weak electrolyte can not be calculated by using λ_m^o for individual ions.

32. Given below are two statements, one is labelled **Assertion A** and the other is labelled as **Reason R.**
Assertion A: Hypophosphorous acid is a reducing agent and converts silver nitrate into silver metal.
Reason R: Hypophosphorous acid contains two P – H bonds.
In the light of the above statements, choose the **correct** answer from the options given below:
(A) A is not correct but R is correct.
(B) Both A and R are correct and R is the correct explanation of A.
(C) Both A and R are correct but R is not the correct explanation of A.
(D) A is correct but R is not correct.

33. Which one of the following metal ions forms blue coloured bead in the borax bead test?
(A) Cr^{3+} (B) Fe^{2+}
(C) Cu^{2+} (D) Ni^{2+}

34. When 300 ml of 0. 2 M H_2A (diprotic strong acid) solution is mixed with 700 ml of 0.1 M KOH solution, the increase in temperature of the final solution is_______× 10^{-3} K.
[Given $H^+(aq) + OH^-(aq) \longrightarrow H_2O \quad \Delta_r H = -13.5\ \text{K cal mol}^{-1}$]
Specific heat of water $= 4.2\ \text{cal K}^{-1}\ \text{g}^{-1}$
Density of water $= 1\ \text{g cm}^{-3}$
Assume there is no change in volume of solution on mixing.
(A) 32
(B) 192
(C) 964
(D) 225

35. Given below are two statements:
Statement-I: Higher the value of K_H at a given temperature, higher the solubility of the gas in a solvent.
Statement-II: The value of K_H increases with increasing temperature.
In the light of above statements, choose the **correct** answer from the options given below:
(A) Statement-I is true but Statement-II is false.
(B) Statement-I is false but Statement-II is true.
(C) Both Statement-I and Statement-II are true.
(D) Both Statement-I and Statement-II are false.

36. When 40 ml of 0.2 M silver nitrate solution is mixed with 20 ml of 0.12 M sodium chromate solution, moles of silver chromate precipitated out is_______
(A) 8×10^{-3}
(B) 2.4×10^{-3}
(C) 4×10^{-3}
(D) 1.2×10^{-3}

37. The **correct** order of metallic radius of Sc, Ti, Cu and Zn is____
(A) Sc > Ti > Cu > Zn
(B) Zn > Cu > Ti > Sc
(C) Sc > Ti > Zn > Cu
(D) Zn > Sc > Ti > Cu

38. Fusion of MnO_2 with KOH in the presence of oxygen produces the dark green solution which on acidification undergoes disproportionation reaction which one of the following statements is **correct** regarding the process____
(A) Green colour is due to MnO_4^{2-} which is paramagnetic.
(B) Green colour is due to MnO_4^- which is diamagnetic.
(C) MnO_4^{2-} undergoes disproportionation in acidic medium to give MnO_4^- and MnO.
(D) Green colour is due to MnO_4^- in which all the Mn – O bond length are identical.

39. Which one of the following statements is **incorrect**?
(A) The actinoids show a greater range of oxidation states than the lanthanoids because the 5f, 6d and 7s levels are of comparable energies.
(B) 5f electrons can participate in bonding to a far greater extent than that 4f orbitals because unlike 4f orbitals, 5f orbitals are not as buried as 4f orbirtals.
(C) The lanthanoid contraction is more important than the actinoid contraction because the chemistry of elements succeeding the actinoids are much less known at the present time.
(D) The magnetic properties of the actinoids are less complex than the lanthanoids.

40. Match List – I with List – II:

List – I		List – II	
(A)	$\left[Mn(H_2O)_6\right]^{2+}$	(I)	Colourless
(B)	$\left[Cr(H_2O)_6\right]^{3+}$	(II)	Green
(C)	$\left[Fe(H_2O)_6\right]^{2+}$	(III)	Purple
(D)	$\left[Sc(H_2O)_6\right]^{3+}$	(IV)	Pink

Choose the **correct** answer from the options given below:
(A) A – III, B – IV, C – I, D – III
(B) A – IV, B – II, C – III, D – I
(C) A – IV, B – III, C – I, D – II
(D) A – IV, B – III, C – II, D – I

41. How many grams of ethylene glycol should be added to 10 litre of water to prevent its freezing at 254.4 K? $\left(K_f \text{ for water is } 1.86 \text{ K Kg mol}^{-1} \text{ and } T_f^o \text{of water is } 273K\right)$
(A) 4200
(B) 5800
(C) 6200
(D) 8200

42. Which one of the following sulphides is **not** soluble in yellow ammonium sulphide?
(A) As_2S_5
(B) SnS_2
(C) CuS
(D) Sb_2S_3

43. Which one of the following statements is **correct?**
(A) $Cr_2O_7^{2-}$ is stronger oxidizing agent than MnO_4^- in acidic medium.
(B) $Cr_2O_7^{2-}$ can act as self indicator in titration of oxalic acid in acidic medium.
(C) Oxygen gas is not produced on heating $KMnO_4$.
(D) Potassium permanganate forms dark purple crystals which are isostructural with those of $KClO_4$.

44. Which one of the following elements is **not** an actinoid?
(A) Er
(B) Es
(C) Fm
(D) Md

45. Which one of the following statements is **incorrect** regarding Cu^{2+}, Cd^{2+}, Pb^{2+} and Bi^{3+} ions?
(A) Only Cd^{2+} and Pb^{2+} ions will give white precipitate of hydroxide on reaction with NaOH.
(B) Only Cd^{2+}, Pb^{2+} and Bi^{3+} ions will give white precipitate of hydroxide on reaction with NaOH.
(C) Only Cu^{2+} will give blue precipitate of hydroxide on reaction with NaOH.
(D) $Pb(OH)_2$ is soluble in excess of NaOH.

46. Choose the **correct** statement among the following?
(A) $\left[FeCl_4\right]^-$ has square planar geometry
(B) $\left[Fe(C_2O_4)_3\right]^{3-}$ has one unpaired electrons
(C) $\left[Co(C_2O_4)_3\right]^{3-}$ has zero unpaired electrons
(D) $\left[Ni(CO)_4\right]$ has square palanar geometry

47. Which one of the following statement is **correct**?
(A) Ammonia gas can not be prepared by hydrolysis of calcium cyanamide.
(B) On heating ammonium nitrite, NH_3 gas is formed.
(C) On heating ammonium dichromate, laughing gas is formed.
(D) On heating ammonium carbonate, ammonia is formed.

48. Bleaching powder contains a salt of an oxoacid as one of its components. The anhydride of that oxoacid is
(A) Cl_2O (B) Cl_2O_7
(C) ClO_2 (D) Cl_2O_6

49. Consider the following phosphorus based oxoacids
$H_4P_2O_6$, $H_3P_3O_9$, $H_4P_2O_8$, $H_4P_2O_5$ and $H_5P_3O_{10}$
Amongst these oxoacids, the number of oxoacids with P – O – P bond is____
(A) 4 (B) 2
(C) 3 (D) 5

50. Upon heating Pb_3O_4 gas Q is formed. Excess amount of Q reacts with white phosphorous to give R. Reaction of R with pure $HClO_4$ gives Y and Z which of the following option is **correct** regarding the above scheme?
(A) R is P_4O_6 (B) Q is O_3
(C) Y and Z are Cl_2O_4 and HPO_3 (D) Y and Z are Cl_2O_7 and HPO_3

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. For the electrochemical cell
$Ca(s)+Ca^{2+}(aq.,1\,M)\,||\,Cu^{2+}(aq.,1\,M)\,|\,Cu(s)$
The standard emf of the cell is 3.20 V at 300 K. When the concentration of Ca^{2+} is changed to x M, the cell potential changes to 3.185 V at 300 K. The value of 100x is__________
[Given: $\frac{F}{R}=11500\ KV^{-1}$, where F is the Faraday constant and R is the gas constant, $\ell n3.15=1.15$]

52. In a 0.2 molal silver nitrate aqueous solution, an equal volume of 0.2 molal aqueous barium chloride solution is added. The difference in the boiling points of water in the original silver nitrate solution and after mixing barium chloride solution in silver nitrate solution is $\mathbf{x}\times 10^{-2}\ ^{o}C$. The **|x|** is_____
[Assume : Densities of silver nitrate solution and solution formed after mixing of barium chloride in the silver nitrate solution are as the same as that of water and soluble salts dissociate completely. Molal elevation constant $K_b = 0.5\ K\,kg\ mol^{-1}$. Boiling point of pure water as $100^{o}C$]

53. When $M_{0.96}O$ is heated in presence of oxygen, it converts to M_2O_3. The n-factor of $M_{0.96}O$ is x, then 100x is ________[M is n element]

54. The reaction of XeF_4 and O_2F_2 at 143 K give a xenon compound (P). The number of mole of HF produced by the complete hydrolysis of 1 mol of P is______

55. Magnesium sulphate heptahydrate is used to fortify foods with magnesium. The amount (in grams) of the salt required to achieve 24 parts per million of magnesium in 100 kg of wheat is______(nearest integer)

56. The total number of unpaired electrons present in $\left[Cr\left(NH_3\right)_6\right]Cl_3$ is____

57. The total number metal ions (aqueous solution) which can form black precipitate with KI (not present in excess) are_____
$Ag^{+}, Pb^{2+}, Hg_2^{2+}, Hg^{2+}, Bi^{3+}, Cu^{2+}, Sb^{3+}$

58. 20 moles of an ideal gas at 300 K in thermal contact with surroundings is compressed isothermally and reversibly from 5 L to 1 L. In this process, the change in entropy of surroundings $\left(\Delta S_{surr}\right)$ in calories K^{-1} is ______$\left[R = 2 \text{ cal mol}^{-1} K^{-1}\right]\left(\ell n5 = 1.6\right)$

59. The total number of metal ions (aqueous solution) which can form soluble complex with excess of NH_4OH solution are____
$Pb^{2+}, Cu^{2+}, Ag^{+}, Hg^{2+}, Zn^{2+}, Ni^{2+}, Bi^{3+}, Cd^{2+}, Al^{3+}, Fe^{3+}, Mg^{2+}$

60. In Duma's method for estimation of nitrogen, 0.1 gram an organic compound gave 76 ml of nitrogen collected at 700 K and 714 mm of Hg pressure. The percentage composition of nitrogen in the compound is_____[Round off the nearest integer] [Given aqueous tension at 700 K is 14 mm of Hg] $\left[R = 0.08 \text{ L atm mol}^{-1} K^{-1}\right]$

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If S_n is sum of n terms of the series 2 + 4 + 7 + 11 + 16 + …, then $\left(\lim\limits_{n\to\infty}\frac{S_n}{n^3}\right) =$

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$
(C) 3 (D) 6

62. If $\begin{bmatrix} 1 & 2 & \alpha \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 22 & 2024 \\ 0 & 1 & 44 \\ 0 & 0 & 1 \end{bmatrix}$, then n + α is equal to

(A) 144 (B) 155
(C) 181 (D) 191

63. If l, m, n, x, y, z are real and $l^2 + m^2 + n^2 = 36$, $x^2 + y^2 + z^2 = 49$ and lx + my + nz = 42 then $\frac{x+y+z}{\ell+m+n}$ is equal to

(A) $\frac{1}{7}$ (B) $\frac{1}{6}$
(C) 1 (D) $\frac{7}{6}$

64. The mean of 5 observation is 4 and their variance is 9.2 If three of the observation are 1, 2 and 6 then other two observation can be

(A) 2 and 9 (B) 3 and 8
(C) 4 and 7 (D) 5 and 6

65. Let α, β roots of equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, ($n \in N$) then the value of $(aS_{2023} + bS_{2022} + c \,.\, S_{2021})$ is equal to

(A) abc (B) a + b + c
(C) c (D) 0

66. If α, β be two roots of the equation $x^2 + (24)^{1/4}x + 6^{1/2} = 0$, then $\alpha^8 + \beta^8$ is equal to

(A) 12 (B) 144
(C) 72 (D) 228

67. If 1, α_1, α_2, …, α_{2022} are $(2023)^{th}$ roots of unity then the value of $\sum\limits_{r=1}^{2022} r\left(\alpha_r + \alpha_{2023-r}\right)$ equals

(A) – 2023 (B) 2023
(C) 2022 (D) 0

68. A five letter word is to be formed such that the letters appearing in the odd position are taken from the unrepeated letters of the word FIITJEELIMITED, whereas the letters which occupy even places are taken from amongst the repeated letters, then the total number of such words are
(A) 390 (B) 360
(C) 540 (D) 450

69. Let x be the number of times tail occurs in n tosses of a fair coin. If P(x = 4), P(x = 5) and P(x = 6) are in A.P. (where P(x = r) denotes probability of r tails in n tosses) then the value of n is divisible by
(A) 2 (B) 4
(C) 5 (D) 7

70. If α and β be the roots of the equation $x^2 + kx - \frac{1}{2k^2} = 0$, where $k \in R$. Then the maximum value of $\alpha^4 + \beta^4$ is
(A) $4\sqrt{2}$ (B) $2 - 2\sqrt{2}$
(C) $2 + \sqrt{2}$ (D) 4

71. The complex number satisfy equations $\log_5(|z| + 3) - \log_{\sqrt{5}}(|z| - 1) = 1$ and arg(z – 1 – i) = $\frac{\pi}{4}$ is
(A) 1 + i (B) $\sqrt{2}(1+i)$
(C) $\sqrt{3}(1+i)$ (D) none of these

72. If z_1 and z_2 are two now zero complex numbers which satisfies $z_1\bar{z}_1 + z_1\bar{z}_1 z_2 = z_2\bar{z}_2 + z_2\bar{z}_2 z_1$ then $\frac{z_2 - \bar{z}_2 + \bar{z}_1 - z_1}{\bar{z}_2 z_1 - z_2\bar{z}_1}$ is equal to (where $\bar{z}_2 z_1 - z_2\bar{z}_1 \neq 0$)
(A) 0 (B) 1
(C) 2 (D) 1/2

73. If the system of equation 2x – y + z = 0, x – 2y + z = 0, tx – y + 2z = 0 has infinitely many solutions and f(x) be continuous function such that f(x + 5) +f(x) = 2 then $\int_0^{-2t} f(x)dx$ is equal to
(A) – 10 (B) – 5
(C) 0 (D) 5

74. Let $S = \sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}} + \sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}} + ... + \sqrt{1+\frac{1}{(2023)^2}+\frac{1}{(2024)^2}}$, then $|2024(S - 2023)|$ is equal to
(A) 2024 (B) 2023
(C) 2 (D) 1

75. A line L_1 passes through a point with position vector $\vec{a}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and is parallel to $\vec{b}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$. Another line L_2 passes through a point with position vector $\vec{a}_2 = 3\hat{i} + 4\hat{j} + \hat{k}$ and is parallel to $\vec{b}_2 = 4\hat{i} + \hat{j} + 3\hat{k}$. Equation of plane equidistant from L_1 and L_2 is
(A) $\vec{r} \cdot (5\hat{i} + 10\hat{j} - 10\hat{k}) = \frac{9}{2}$ (B) $\vec{r} \cdot (5\hat{i} + 10\hat{j} - 10\hat{k}) = 25$
(C) $\vec{r} \cdot (\hat{i} + 2\hat{j} - 2\hat{k}) = \frac{9}{2}$ (D) $\vec{r} \cdot (\hat{i} + 2\hat{j} - 2\hat{k}) = 9$

76. If $(1 + x + x^2 + x^3)^n = \sum_{r=0}^{3n} a_r \cdot x^r$ then $\sum_{r=0}^{3n} r \cdot a_r$

(A) $3n \cdot 4^n$ (B) $3n \cdot 4^{n-1}$

(C) $6n \cdot 4^n$ (D) $6n \cdot 4^{n-1}$

77. A sphere is inscribed in a tetrahedron whose vertices are (8, 0, 0), (0, 6, 0), (0, 0, 2) and (0, 0, 0). If radius of sphere is m/n (where m and n are relatively prime), then m + n is equal to

(A) 4 (B) 5

(C) 6 (D) 7

78. In an examination consisting of four parts A, B, C and D each part containing 30 questions. If a student randomly attempt the exam, then the probability that he attempts atleast 60 questions in the exam is

(A) $\frac{1}{2}+\frac{{}^{120}C_{60}}{(2)^{121}}$ (B) $\frac{1}{2}-\frac{{}^{120}C_{60}}{(2)^{121}}$

(C) $\frac{1}{2}+\frac{{}^{120}C_{60}}{(2)^{120}}$ (D) $\frac{1}{2}-\frac{{}^{120}C_{60}}{(2)^{120}}$

79. If n positive integers are taken at random and multiplied together, then the probability that last digit of the product is 2, 4, 6 or 8 is

(A) $\left(\frac{4}{5}\right)^n$ (B) $\left(\frac{4}{5}\right)^n-\left(\frac{0}{5}\right)^n$

(C) $\left(\frac{1}{5}\right)^n$ (D) $\left(\frac{2}{5}\right)^n$

80. The length of perpendicular from the point (5, 5, 5) to the plane containing the lines $\frac{x-4}{1}=\frac{y+1}{-2}=z$ and $4x-y+5z-7=0$ = 2x – 5y – z – 3 is

(A) 14 (B) $14\sqrt{2}$

(C) $2\sqrt{14}$ (D) $\frac{15}{7}\left(\sqrt{14}\right)$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let image of line 7x = 3 – 6t, y = 2t, z = 3 + 2t in plane 3x + 4y – 5z + 26 = 0 be L and L is perpendicular to the line $\frac{x-4}{-1}=\frac{y-5}{k}=\frac{z-6}{-3}$ then k is equal to _________

82. The value of $\left(\frac{C_1}{C_0}+2\cdot\frac{C_2}{C_1}+3\cdot\frac{C_3}{C_2}+...+20\frac{C_{20}}{C_{19}}\right)$ is _________ (where $C_r = {}^{20}C_r$)

83. If origin O lines inside the ΔABC. Such that $\overrightarrow{OA}+\alpha\overrightarrow{OB}+\beta\overrightarrow{OC}=0$ and if area of ΔABC = 5 area of ΔOBC, then $\alpha + \beta$ = _________

84. $\alpha, \beta, \gamma, \delta$ are roots of $f(x) = 5x^4 + ax^3 + bx^2 + cx + d$ and α, γ, δ are roots of $g(x)=x^3 - 9x^2 + kx - 24$ also α, γ, δ are in A.P. and $\alpha, \beta, \gamma, \delta$ are in H.P. (where $\alpha < \beta < \gamma < \delta$), then the value of f(5) . g(5) is equal to _________

85. Let A be a square matrix of order 3 whose elements are real numbers and adj(adj(adj A)) = $\begin{bmatrix} 27 & 0 & -3 \\ 0 & 9 & 0 \\ 0 & 3 & 27 \end{bmatrix}$ then the value of $3\left(\mathrm{Tr}\left(A^{-1}\right)\right)$ is equal to _________ (adj A and Tr(A) denotes adjoint matrix and trace of matrix A respectively)

86. If a sequence $a_1, a_2, a_3, \ldots, a_{100}$ satisfies the relation $a_{n+1} = a_n + \lambda$ (where λ is real constant) for n = 1, 2, …., 99 such that $a_9 = 15$ then value of $S_{12} - S_5$ is equal to _________

87. Let $1 - x + x^2 - x^3 + \ldots + x^{22} - x^{23} = a_0 + a_1(1 + x) + a_2(1 + x)^2 + \ldots + a_{23}(1 + x)^{23}$ then a_2 =_____

88. If f(x) be a polynomial of degree 4 with leading coefficient 1 satisfies f(1) = 5, f(2) = 10, f(3) = 15 then the value of f(7) + f(–3) is _________

89. Number of ordered pair(s) (a, b) of real numbers such that $(a + ib)^{2022} = a - ib$ holds good, is____

90. The term independent of x is $(1 + x + x^{-2} + x^{-3})^{10}$ is _________